

SUPPLEMENTARY MATERIAL

Adaptive Nonlinear Vector Autoregression: Robust Forecasting for Noisy Chaotic Time Series

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This document provides supplementary materials that support and extend the main manuscript. It includes:

- **Supplementary Example** — A full demonstration of the Adaptive Nonlinear Vector Autoregression (Adaptive NVAR) workflow, including data preprocessing, model construction, and prediction on a sample chaotic system.
- **Supplementary Tables 3–7** — Comprehensive tabulations of:
 - **Table 3–5** — Optimal hyperparameter configurations selected via validation RMSE for each dynamical system and noise level.
 - **Table 6** — Fixed experimental settings such as time-series length, washout steps, and input/output scaling factors.
 - **Table 7** — Feature dimensionality for each delay embedding and polynomial expansion used in the NVAR and Adaptive NVAR models.
- **Supplementary Figures 1–12** — Forecasting performance visualizations across multiple chaotic systems:
 - **Figures 1–3** — Predictions on the test data under 0% noise.
 - **Figures 4–6** — Predictions on the test data under 10% noise.
 - **Figures 7–9** — Predictions on the test data under 20% noise.
 - **Figures 10–12** — Predictions on the test data under 30% noise.
- **Additional Methodological Notes** — Note on hyperparameter search strategies for the Adaptive NVAR

Supplementary Example

We illustrate our adaptive NVAR model using a simple 3-dimensional time series with delay embedding and joint feature learning.

- (i) Dimension of the state vector: $d = 3$.
- (ii) Number of delays: $k = 2$.
- (iii) Neural network output dimension: $d_{\mathcal{NN},i} = dk(dk + 1)/2$, to match the number of fixed nonlinear features in NVAR. Alternatively, $d_{\mathcal{NN},i} = cdk$, with c being a tunable constant.
- (iv) Input dimension: $dk + d_{\mathcal{NN},i} = 6 + 21 = 27$.
- (v) Number of training points: $n = 5$ ($I_{\text{train}} = \{2, 3, 4, 5, 6\}$ are valid training indices for $k = 2$).
- (vi) For each $i \in I_{\text{train}}$, we use $H_{\text{lin},i} = X_i \oplus X_{i-1} \in \mathbb{R}^6$.

Let $X_i \in \mathbb{R}^d$ for $i = \{1, \dots, 10\}$ be our toy series dataset, where $d = 3$. The following table lists the components of X_i .

i	1	2	3	4	5	6	7	8	9	10
$x_{1,i}$	0.1	0.2	0.3	0.5	0.4	0.7	0.6	0.8	0.9	0.5
$x_{2,i}$	0.3	0.1	0.6	0.2	0.5	0.1	0.4	0.6	0.5	0.2
$x_{3,i}$	0.2	0.4	0.1	0.3	0.6	0.2	0.3	0.2	0.9	0.2

Table 1: Values of $x_{j,i}$ for $j = 1, 2, 3$ and $i = 1, \dots, 10$.

We now construct the linear feature vectors and their corresponding targets for training:

i	$H_{\text{lin},i}$	Target X_{i+1}
2	$X_2 \oplus X_1$	X_3
3	$X_3 \oplus X_2$	X_4
4	$X_4 \oplus X_3$	X_5
5	$X_5 \oplus X_4$	X_6
6	$X_6 \oplus X_5$	X_7

Table 2: Linear combination inputs $H_{\text{lin},i}$ and their target outputs X_{i+1} .

The feature vectors are then in the following vector spaces:

1. The linear feature vector $H_{\text{lin},i} \in \mathbb{R}^{dk} = \mathbb{R}^6$.
2. The neural network (nonlinear) feature vector $H_{\mathcal{NN},i} = \mathcal{NN}(H_{\text{lin},i}; \theta) \in \mathbb{R}^{d_{\mathcal{NN},i}} = \mathbb{R}^{21}$.
3. The total feature vector, defined by concatenating the linear and nonlinear features $H_{\text{total},i} = H_{\text{lin},i} \oplus H_{\mathcal{NN},i} \in \mathbb{R}^{dk+d_{\mathcal{NN},i}} = \mathbb{R}^{27}$.

The neural network $\mathcal{NN}(H_{\text{lin},i}; \theta)$ is defined as a simple MLP with one hidden layer and a tanh activation function. Let the hidden layer have $h = 24$ units. The MLP transformation is given by:

$$H_{\mathcal{NN},i} = \mathcal{NN}(H_{\text{lin},i}; \theta) = W \cdot \tanh(W_{\text{in}} H_{\text{lin},i}),$$

where $W_{\text{in}} \in \mathbb{R}^{h \times (dk)} = \mathbb{R}^{24 \times 6}$ and $W \in \mathbb{R}^{d_{\mathcal{NN},i} \times h} = \mathbb{R}^{21 \times 24}$.

After concatenating $H_{\text{lin},i}$ and $H_{\mathcal{NN},i}$ into $H_{\text{total},i}$, we perform the readout layer as follows:

$$\hat{Y}_i = \hat{X}_{i+1} - \hat{X}_i = W_{\text{out}} H_{\text{total},i}.$$

Thus, the parameter set θ is $\theta = \{W_{\text{in}}, W, W_{\text{out}}\}$.

The mean squared error loss

$$L(\theta, W_{\text{out}}) = \frac{1}{5} \sum_{i=2}^6 \|W_{\text{out}} H_{\text{total},i} - Y_i\|_2^2 = \frac{1}{5} \sum_{i=2}^6 \|W_{\text{out}} [H_{\text{lin},i} \oplus \mathcal{NN}(H_{\text{lin},i}; \theta)] - Y_i\|_2^2$$

is used to compute the gradients with respect to both θ and W_{out} .

After completing training, we evaluate the model using the remaining points, referred to as test points. For each prediction, we use the trained W_{out} and $H_{\text{total},i}$.

$$\hat{Y}_7 = W_{\text{out}} H_{\text{total},7}, \text{ where } H_{\text{total},7} = H_{\text{lin},7} \oplus H_{\mathcal{NN},7}$$

$$\hat{Y}_8 = W_{\text{out}} H_{\text{total},8}, \text{ where } H_{\text{total},8} = H_{\text{lin},8} \oplus H_{\mathcal{NN},8}$$

and so on.

Supplementary Tables

Hyperparameter	0% Noise	10% Noise	20% Noise	30% Noise
<i>Standard NVAR</i>				
Delay embedding length k	50	50	50	50
Ridge regularization λ	10^0	10^{-1}	10^{-2}	10^{-7}
<i>Adaptive NVAR</i>				
Delay embedding length k	30	30	30	50
Number of hidden units in MLP	50	20	10	10
Initial Adam learning rate	10^{-2}	10^{-2}	10^{-4}	10^{-3}
Initial L-BFGS learning rate	0.5	1.0	1.0	1.0
<i>Echo State Network (ESN)</i>				
Input weight scale σ_{in}	0.8	0.8	0.8	0.8
Reservoir spectral radius ρ	1.25	1.55	1.55	1.55
Leaking rate α	0.6	0.3	0.3	0.3
Ridge regularization λ	10^{-2}	10^{-2}	10^{-2}	10^{-2}
Connection probability pr	0.2	0.2	0.2	0.2
<i>Hybrid Echo State Network (HESN)</i>				
Input weight scale σ_{in}	0.8	0.8	0.5	0.5
Knowledge-based input weight scale σ_{kb}	0.02	0.02	0.02	0.02
Reservoir spectral radius ρ	1.55	1.15	1.15	1.55
Leaking rate α	0.4	0.3	0.3	0.4
Ridge regularization λ	10^{-2}	10^{-2}	10^{-6}	10^{-3}
Connection probability pr	0.15	0.15	0.1	0.02

Table 3: Optimal hyperparameters for each model on the Mackey–Glass system under different noise levels.

Hyperparameter	0% Noise	10% Noise	20% Noise	30% Noise
<i>Standard NVAR</i>				
Delay embedding length k	2	50	50	30
Ridge regularization λ	10^0	10^2	10^3	10^1
<i>Adaptive NVAR</i>				
Delay embedding length k	50	50	30	50
Number of hidden units in MLP	500	100	500	1000
Initial Adam learning rate	10^{-8}	10^{-8}	10^{-8}	10^{-4}
Initial L-BFGS learning rate	1.0	0.1	0.1	0.1
<i>Echo State Network (ESN)</i>				
Input weight scale σ_{in}	0.5	0.5	0.5	0.5
Reservoir spectral radius ρ	0.99	0.85	0.9	0.9
Leaking rate α	0.4	0.3	0.2	0.2
Ridge regularization λ	10^{-3}	10^{-3}	10^{-2}	10^{-2}
Connection probability pr	0.05	0.15	0.2	0.1
<i>Hybrid Echo State Network (HESN)</i>				
Input weight scale σ_{in}	0.8	0.5	0.8	0.5
Knowledge-based input weight scale σ_{kb}	0.02	0.02	0.05	0.02
Reservoir spectral radius ρ	0.99	0.99	0.8	0.85
Leaking rate α	0.2	0.5	0.2	0.2
Ridge regularization λ	10^{-2}	10^{-2}	10^{-1}	10^{-1}
Connection probability pr	0.2	0.1	0.2	0.2

Table 4: Optimal hyperparameters for each model on the Lorenz-63 system under different noise levels.

Hyperparameter	0% Noise	10% Noise	20% Noise	30% Noise
Delay embedding length k	10	10	10	10
Number of hidden units in MLP	1000	1000	1000	1000
Initial Adam learning rate	10^{-5}	10^{-5}	10^{-5}	10^{-5}

Table 5: Optimal hyperparameters for Adaptive NVAR on the Lorenz-96 system ($N = 100$) under different noise levels.

Model	Hyperparameter	Value
Standard NVAR	Skipping parameter (s)	1
Adaptive NVAR	Skipping parameter (s)	1
Adaptive NVAR (Lorenz-96)	Mini-batch size	1024
ESN	Washout length (test/validation)	1000 (MG), 500 (L63)
ESN	Number of reservoir neurons	100
HESN	Washout length (test/validation)	1000 (MG), 500 (L63)
HESN	Number of reservoir neurons	100

Table 6: Fixed hyperparameters used in each model.

Model	System	Feature Formula	Parameter Values	Feature Size (m)
Standard NVAR	Mackey-Glass	$dk(dk + 1)/2$	$d = 1, k = 10$	55
Standard NVAR	Lorenz-63	$dk(dk + 1)/2$	$d = 3, k = 10$	4,515
Standard NVAR	Lorenz-96	$dk(dk + 1)/2$	$d = 100, k = 20$	2,001,000
Adaptive NVAR	Mackey-Glass	$dk(dk + 1)/2$	$d = 1, k = 10$	55
Adaptive NVAR	Lorenz-63	$dk(dk + 1)/2$	$d = 3, k = 10$	4,515
Adaptive NVAR	Lorenz-96	$n dk$	$n = 10, d = 100, k = 20$	20,000

Table 7: Examples: Nonlinear feature dimensionality for the NVAR and Adaptive NVAR models. Here, d is the system dimension, k the number of delay taps, and n the number of hidden neurons per delayed variable in the Adaptive NVAR. For high-dimensional systems such as Lorenz-96, the Adaptive NVAR replaces the polynomial feature expansion with a shallow MLP of output size $m = n dk$.

Supplementary Figures

Noise-Free (0%)

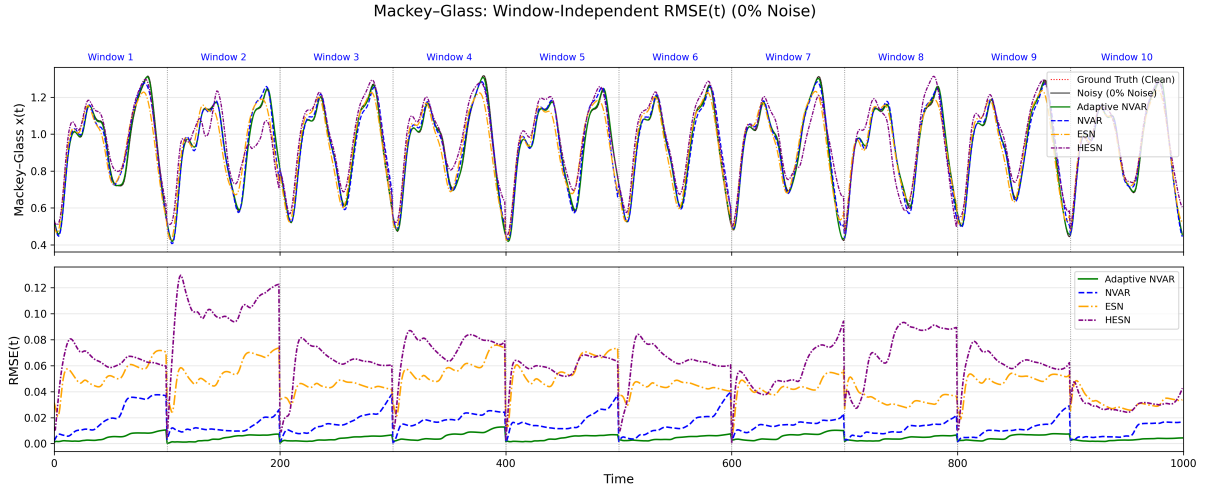


Figure 1: **Forecasting performance of the Mackey–Glass system (0% noise).** Top panel shows the trajectories of the true signal (red, dotted), noisy input (black), and model predictions (colored lines). Bottom panel shows the window-wise RMSE(t) for non-overlapping windows of length 100.

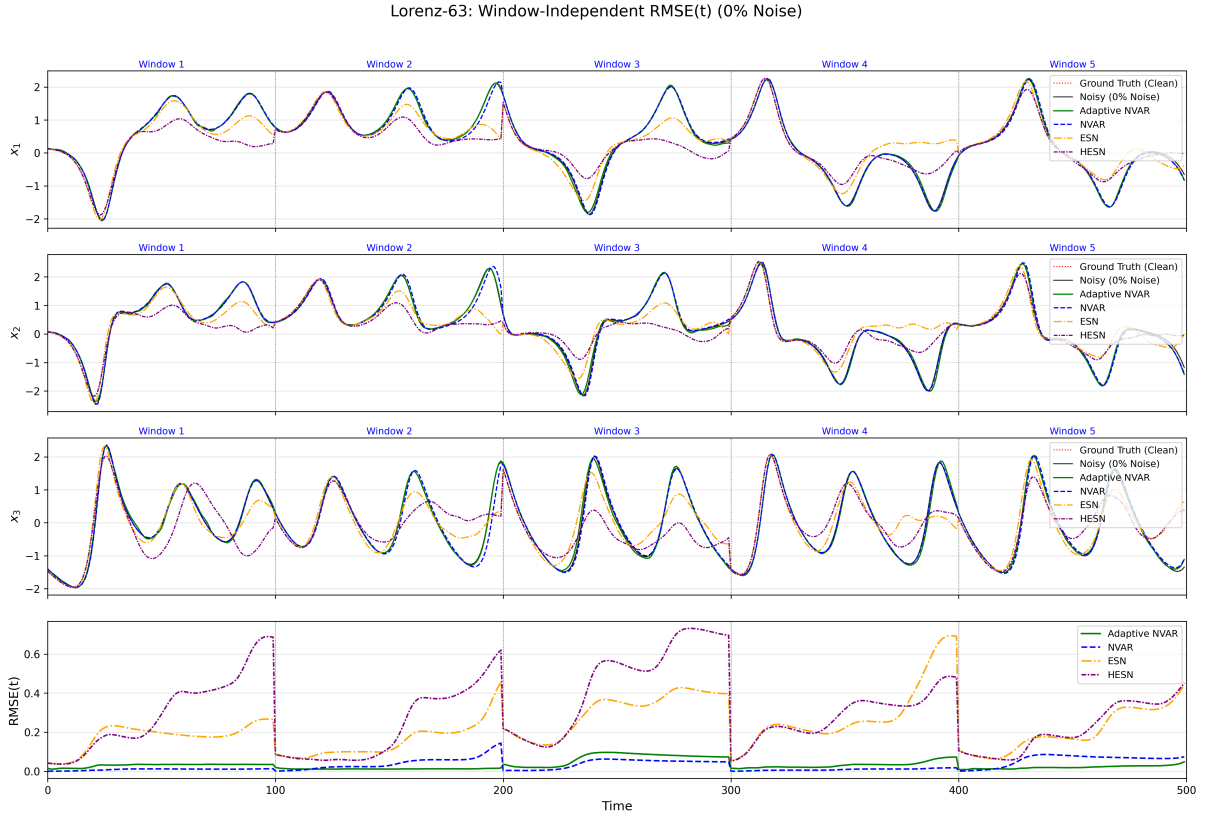


Figure 2: **Forecasting performance of the Lorenz–63 system (0% noise).** Each subplot shows the true signal (red, dotted), noisy input (black), and predictions (colored lines) for each state variable. Vertical dotted lines indicate forecast windows of 100 time steps used to compute window-wise RMSE.

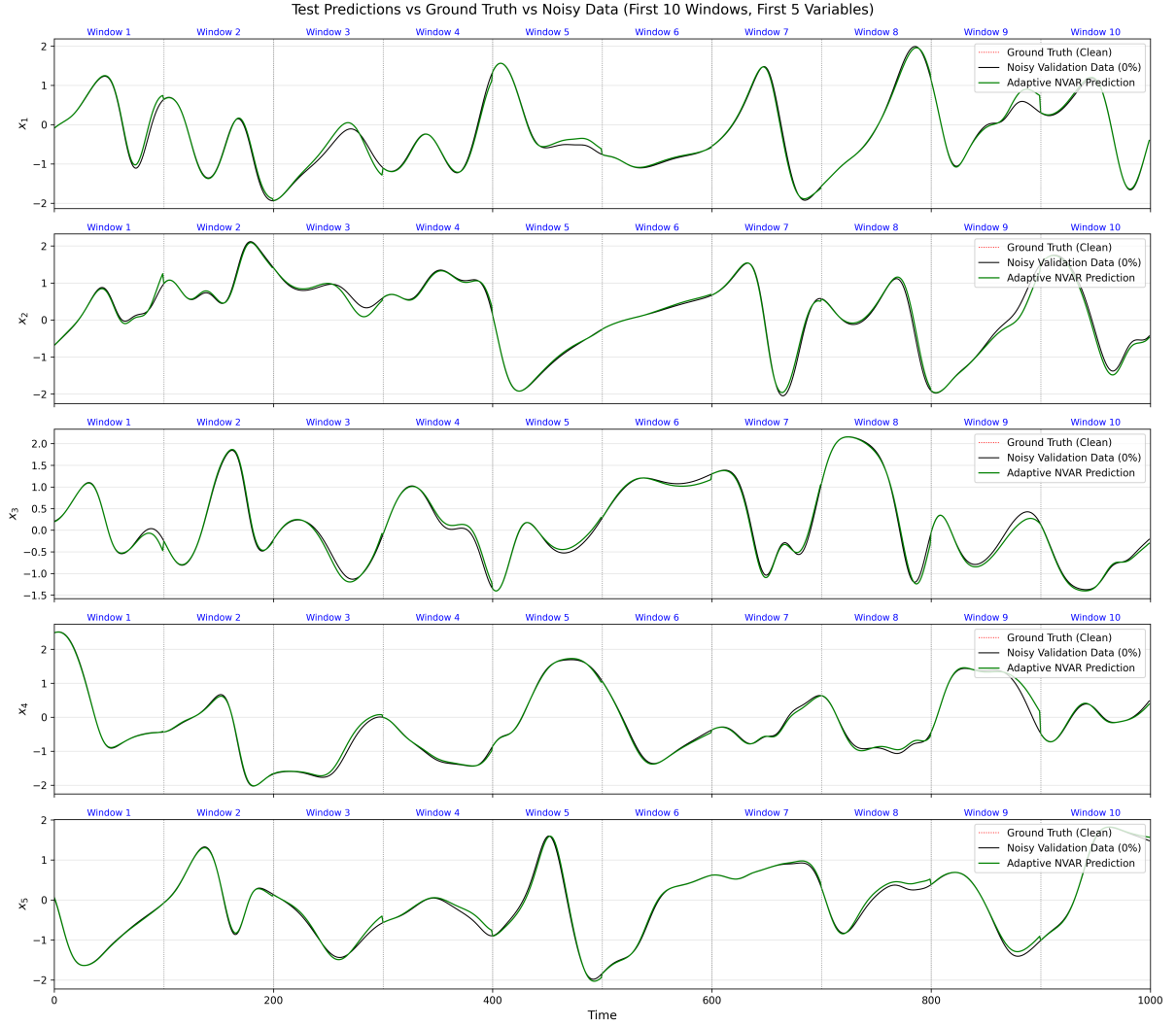


Figure 3: **Forecasting performance of the high-dimensional Lorenz-96 system (0% noise).** True signal (red, dotted), noisy input (black), and Adaptive NVAR predictions (green) for the first 5 variables and first 10 forecast windows (each of 100 time steps).

Low Noise (10%)

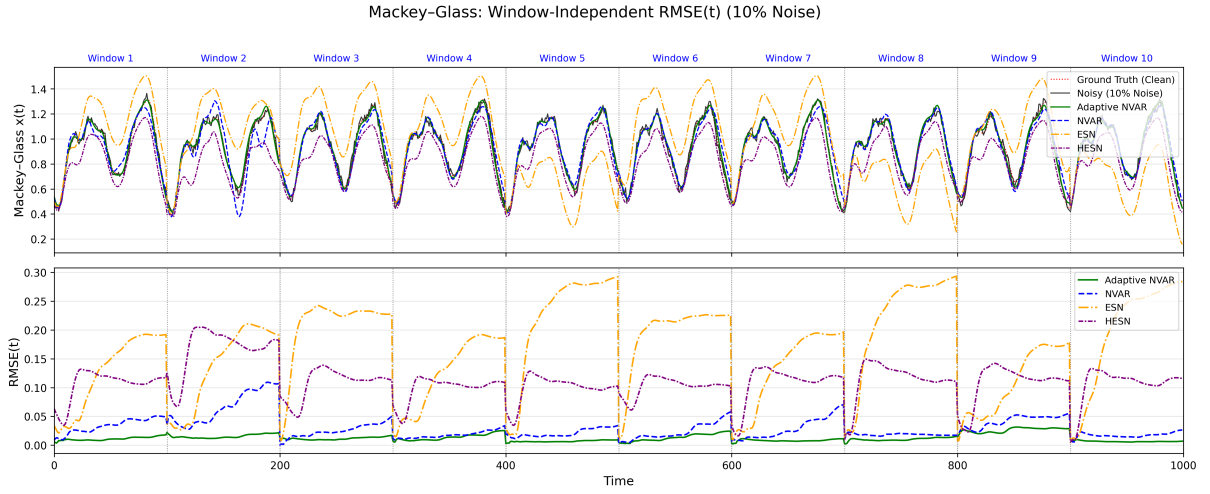


Figure 4: Forecasting performance of the Mackey–Glass system (10% noise).

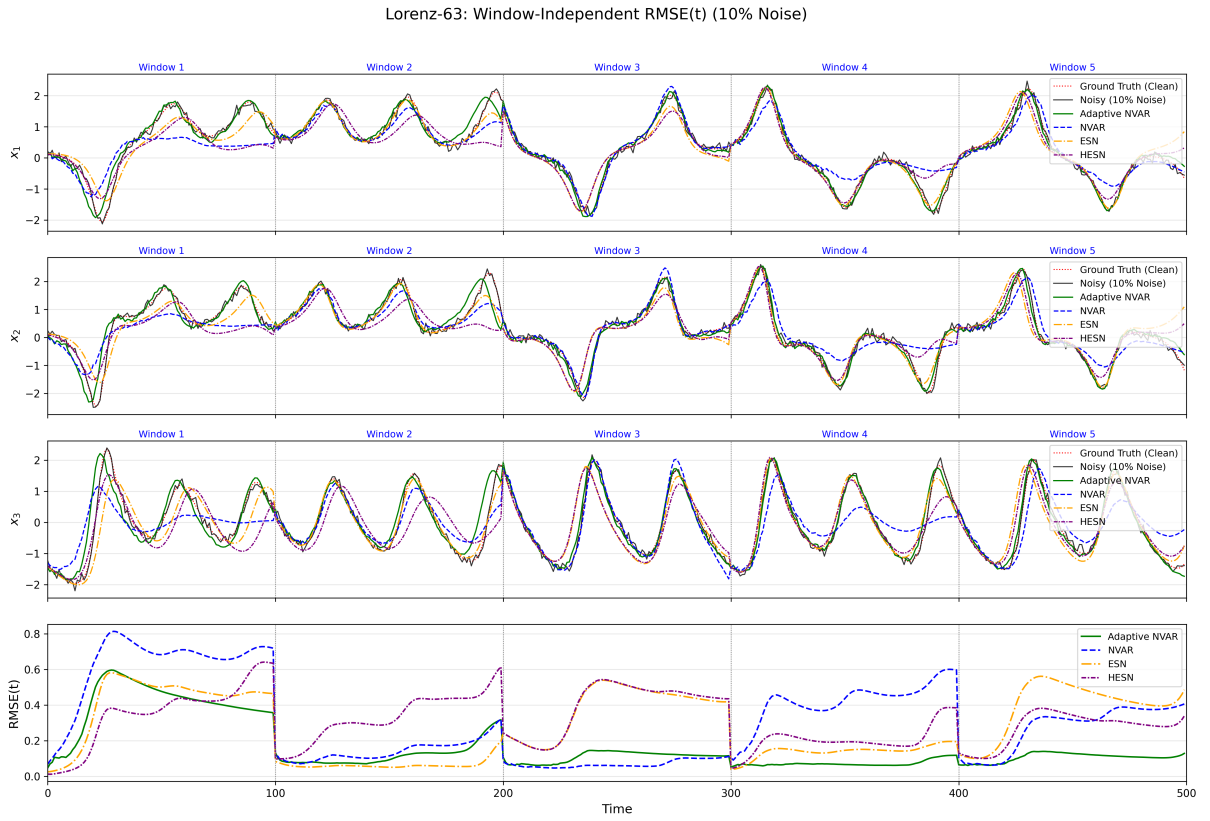


Figure 5: Forecasting performance of the Lorenz–63 system (10% noise).

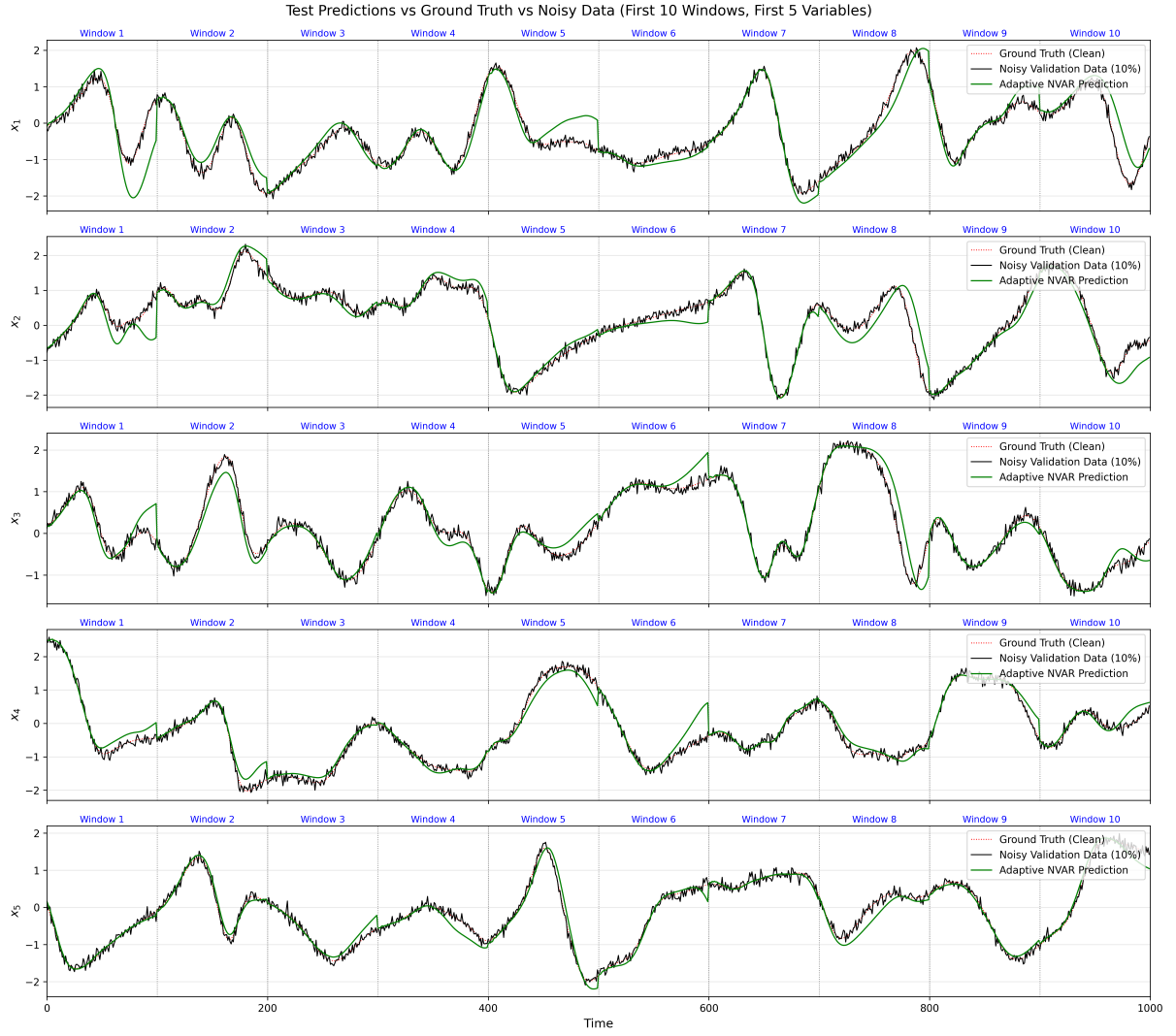


Figure 6: **Forecasting performance of the high-dimensional Lorenz-96 system (10% noise).**

Moderate Noise (20%)

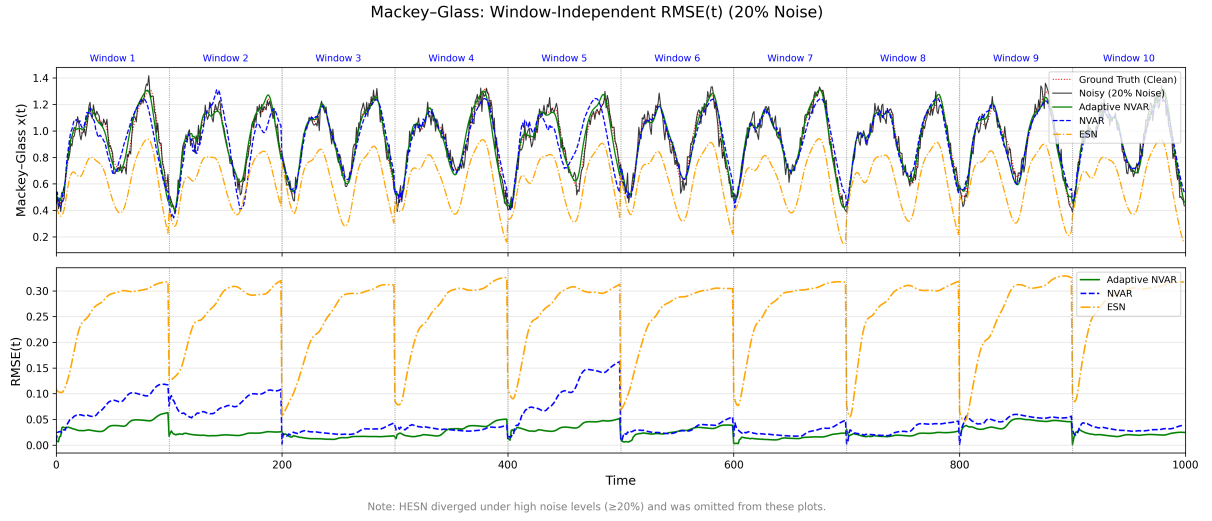


Figure 7: Forecasting performance of the Mackey–Glass system (20% noise).

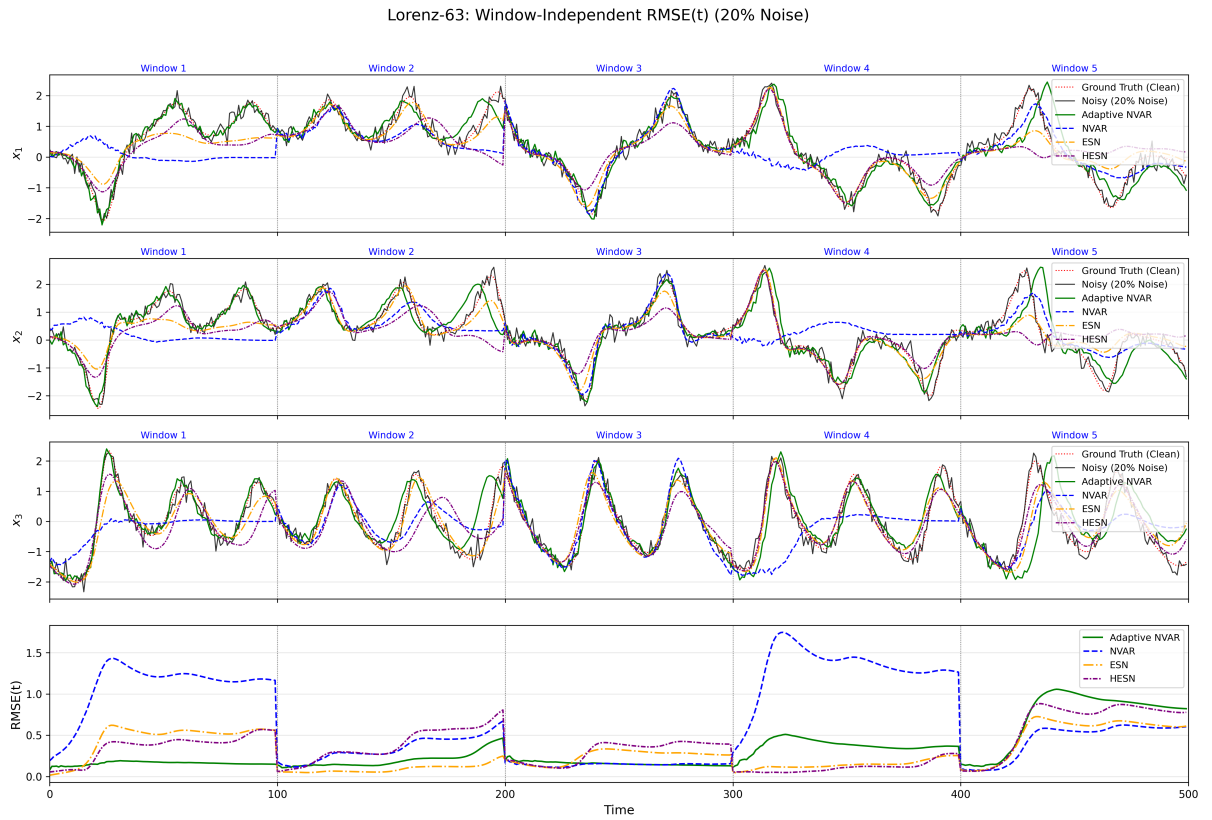


Figure 8: Forecasting performance of the Lorenz–63 system (20% noise).

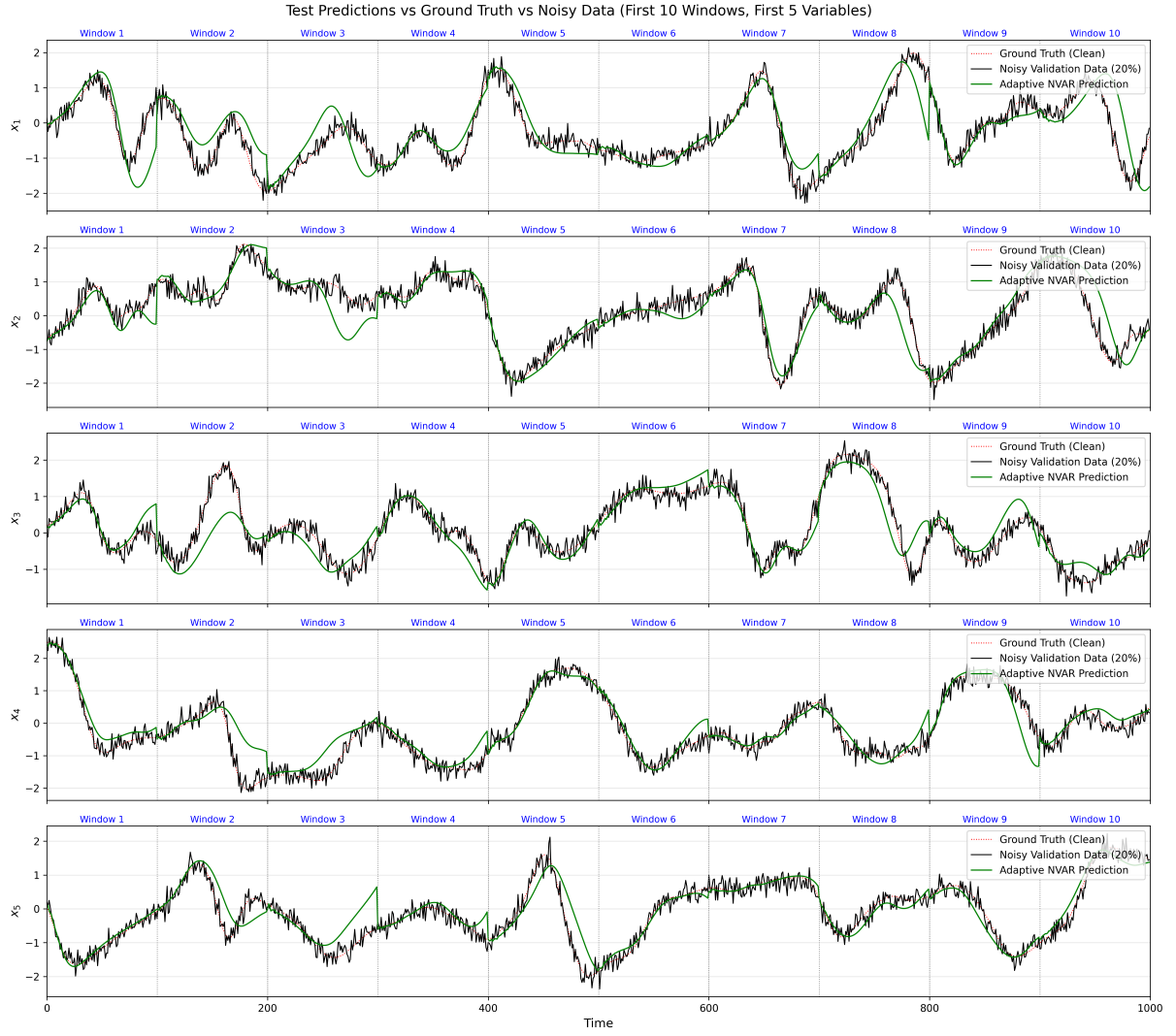


Figure 9: **Forecasting performance of the high-dimensional Lorenz-96 system (20% noise).**

High Noise (30%)

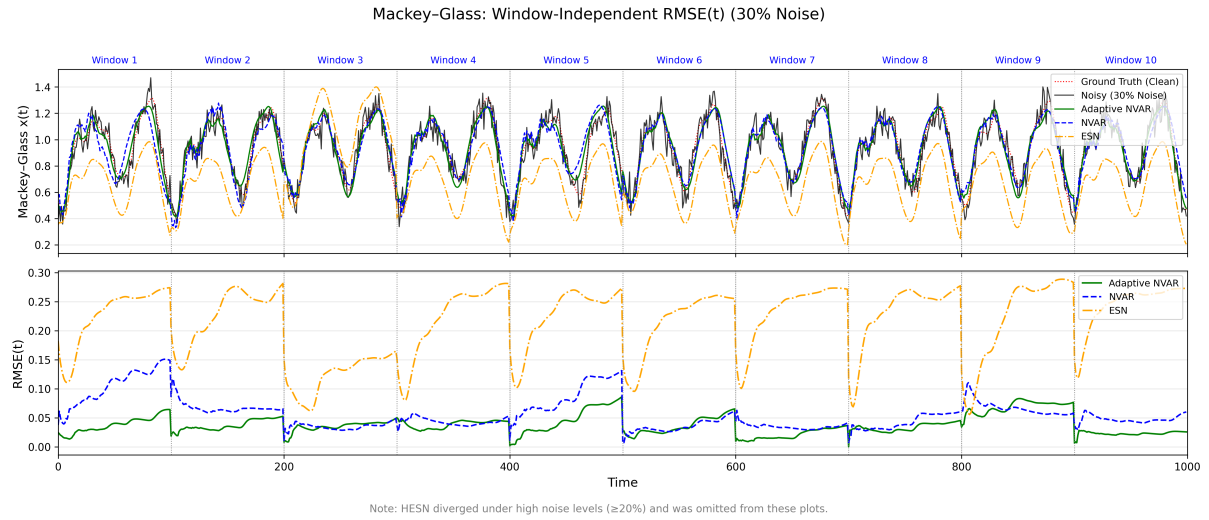


Figure 10: Forecasting performance of the Mackey–Glass system (30% noise).

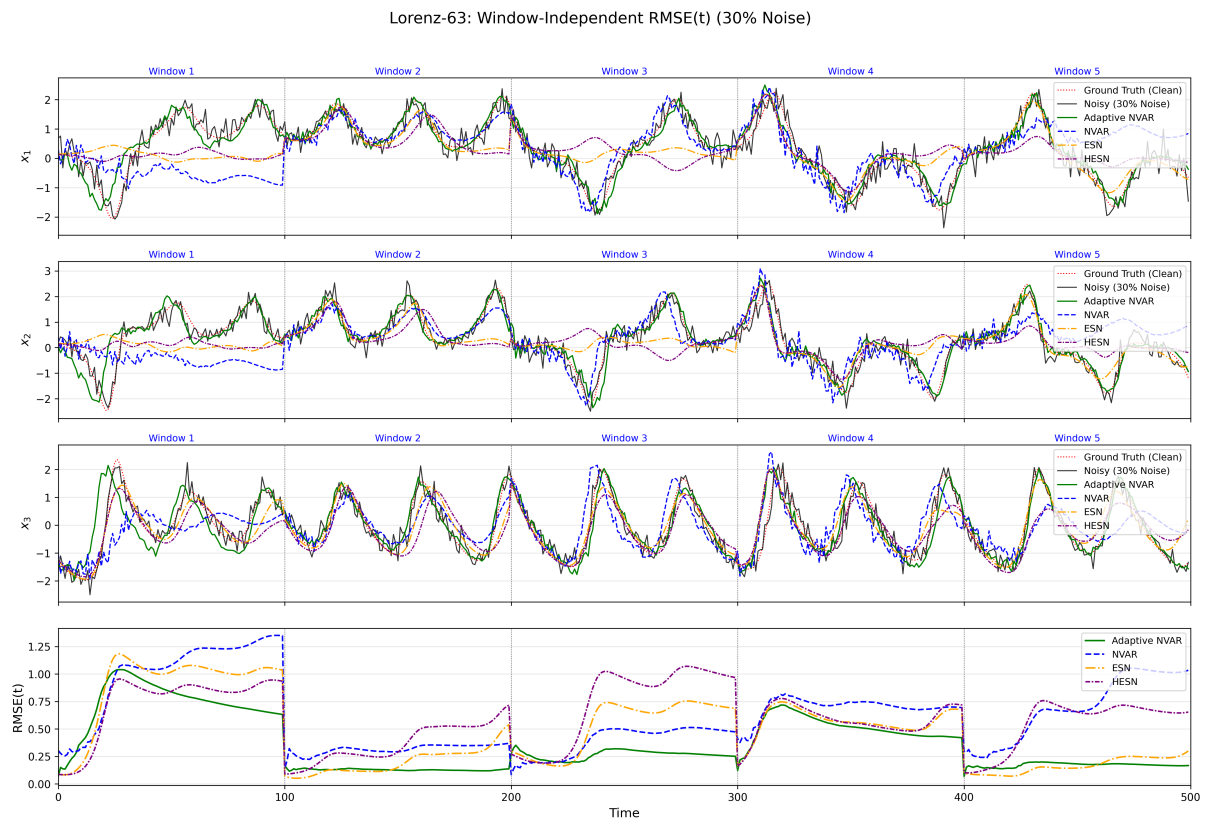


Figure 11: Forecasting performance of the Lorenz–63 system (30% noise).

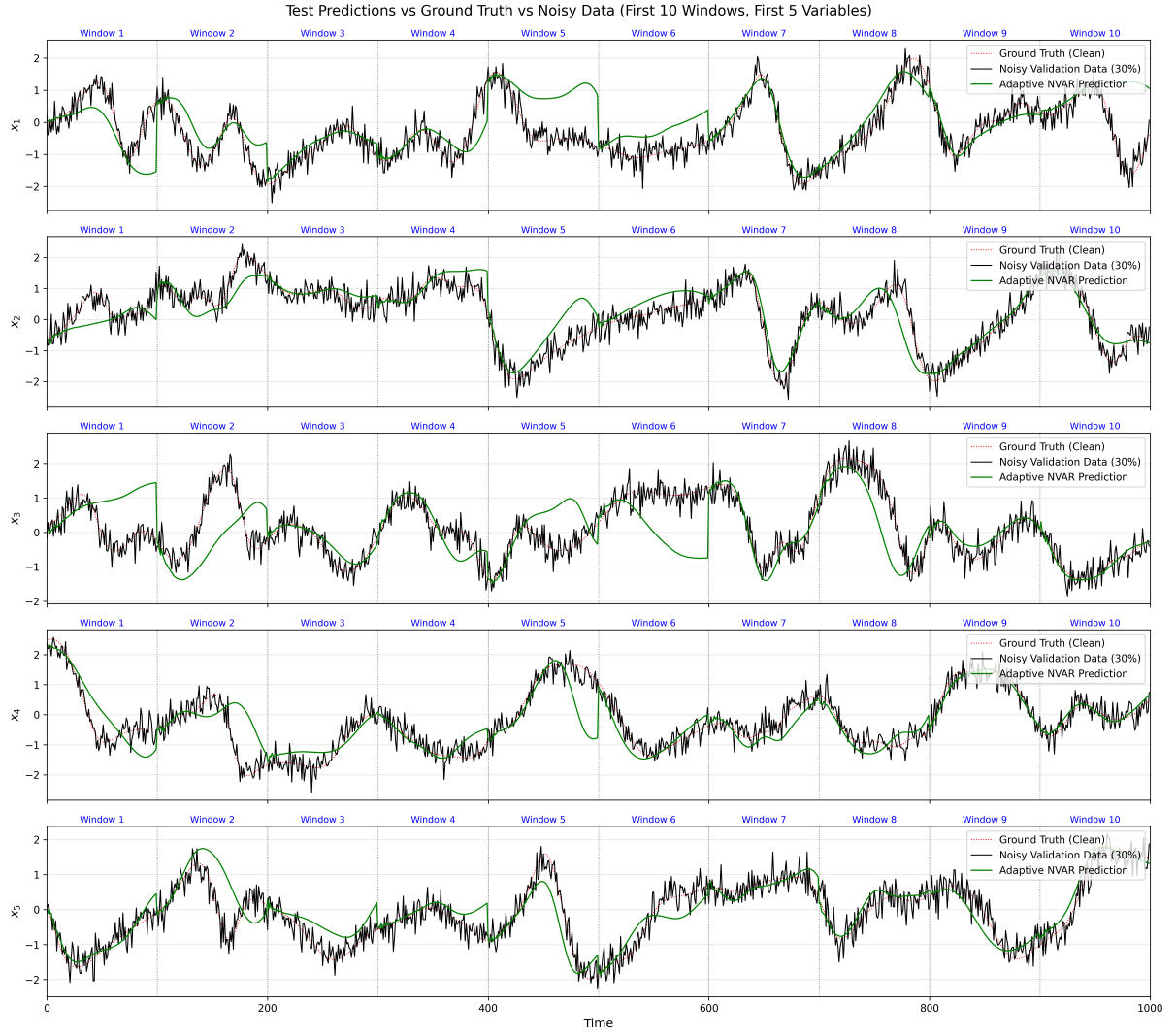


Figure 12: Forecasting performance of the high-dimensional Lorenz-96 system (30% noise).

Note on hyperparameter search strategies

The following are the hyperparameter search intervals listed in Table 1 of the paper.

Method	Parameter	Values
Standard NVAR	Delay parameter (k)	$\{2, 10, 30, 50\}$
	Regularization (λ)	$\{10^{-7}, 10^{-6}, \dots, 10^3, 10^4\}$
Adaptive NVAR	Delay parameter (k)	$\{2, 10, 30, 50\}$
	Hidden units	$\{10, 20, 50, 100, 200, 500, 1000\}$
	Adam learning rate	$\{10^{-10}, 10^{-8}, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$
	L-BFGS learning rate	$\{1.0, 0.5, 0.1, 0.01\}$
ESN	Input weight scale (σ_{in})	$\{0.02, 0.05, 0.10, 0.20, 0.50, 0.80\}$
	Spectral radius (ρ)	$\{0.80, 0.85, 0.90, 0.99, 1.05, 1.15, 1.25, 1.55\}$
	Leaking rate (α)	$\{0.20, 0.30, \dots, 1.00\}$
	Regularization (λ)	$\{10^{-7}, 10^{-6}, \dots, 10^3, 10^4\}$
	Connection probability (pr)	$\{0.01, 0.02, 0.05, 0.10, 0.15, 0.20\}$
HESN	Input weight scale (σ_{in})	$\{0.02, 0.05, 0.10, 0.20, 0.50, 0.80\}$
	Knowledge-based scale (σ_{kb})	$\{0.02, 0.05, 0.10, 0.20, 0.50, 0.80\}$
	Spectral radius (ρ)	$\{0.80, 0.85, 0.90, 0.99, 1.05, 1.15, 1.25, 1.55\}$
	Leaking rate (α)	$\{0.20, 0.30, \dots, 1.00\}$
	Regularization (λ)	$\{10^{-7}, 10^{-6}, \dots, 10^3, 10^4\}$
	Connection probability (pr)	$\{0.01, 0.02, 0.05, 0.10, 0.15, 0.20\}$

Table 8: Grid search ranges for each model’s hyperparameters.

They represent the complete ranges explored across all benchmark systems for all models being evaluated. To ensure computational feasibility and to adapt to the distinct dynamics of each system, system-specific subsets were applied:

- **Mackey–Glass:** Lightweight 1D dynamics allowed a broad exploration of hidden dimensions and learning rates ($k \in \{2, 10, 30, 50\}$, hidden units $\in \{10, 20, 50, 100, 200, 500, 1000\}$, Adam learning rate $\in \{10^{-4}, 10^{-3}, 10^{-2}\}$, L-BFGS learning rate $\in \{1.0, 0.5, 0.1\}$).
- **Lorenz–63:** To capture moderate chaotic complexity, higher hidden dimensions were used for stable optimization ($k \in \{2, 10, 30, 50\}$, hidden units $\in \{100, 500, 1000\}$) and smaller learning rates ($\{10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}\}$).
- **Lorenz–96:** For high-dimensional settings ($d = 100$), computational scalability was the limiting factor. Thus, a reduced grid was employed ($k \in \{2, 10, 30\}$, hidden units $\in \{200, 500, 1000\}$, Adam learning rate $\in \{10^{-3}, 10^{-4}, 10^{-5}\}$).